DMD-AUGMENTED AST GRIDS: A DUAL-STREAM APPROACH FOR PLATE MODAL ANALYSIS

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ABSTRACT

Task B of the 1st DAFx Parameter Estimation Challenge focuses on the blind system identification of plate reverberation. Given an observed displacement impulse response, the objective is to directly extract the underlying acoustic characteristics by estimating a highly dense set of modal frequencies, decay constants, and modal gains within a user-defined frequency band. In this paper, we propose a dual-stream Audio Spectrogram Transformer (AST) grid framework over 10 000 log-spaced frequency bins. The first stream uses a pretrained AST backbone together with a log-energy descriptor to obtain a global embedding of the impulse response. The second stream encodes a four-channel log-frequency grid—log magnitude, normalized Dynamic Mode Decomposition (DMD) decay, normalized DMD gain, and a DMD validity mask—through a one-dimensional convolutional encoder. Both streams are fused into a shared per-bin representation, and three convolutional heads predict mode activity, normalized decay, and normalized gain for each bin. At inference, active bins are selected by thresholding the activity head; frequency, decay, and gain are read from the network outputs.

Index Terms— Plate reverberation, modal analysis, parameter estimation, audio spectrogram transformer, dynamic mode decomposition.

1. INTRODUCTION

Plate reverberation is a classic artificial reverberation technique based on the vibration of a thin plate [1, 2]. Its impulse response is determined by a dense set of resonant modes whose frequencies, decay rates, and modal gains depend on the plate material, geometry, boundary conditions, tension, and observation position [3, 4]. Although forward simulation of plate vibration is well established, estimating modal parameters directly from an observed displacement response remains difficult because the number of modes grows rapidly with frequency and because decay and gain vary over many orders of magnitude.

Task B of the 1st DAFx Parameter Estimation Challenge addresses this dense modal inverse problem. Given a one-second displacement impulse response, the system must output a variable-length list of modes, each described by a natural frequency f_0, a decay constant σ, and a modal gain b, restricted to a user-defined band. The official metric combines clipped per-mode relative errors with a penalty on mode-count mismatch, and estimated modes are matched to ground truth by Hungarian assignment on log-frequency distance. Because thousands of modes may be present in a single response, purely manual or exhaustive search strategies are impractical; challenge baselines therefore rely on signal-processing pipelines such as peak picking in the log-magnitude spectrum and band-wise post-processing of spectral peaks.

Dynamic Mode Decomposition (DMD) offers a complementary route to extracting local modal dynamics from time-series data [5, 6, 7, 8]. Given a sequence of snapshots $\{\mathbf{x}(t_n)\}$, DMD approximates the underlying evolution by a low-rank linear operator whose eigenvalues encode oscillation frequency and decay rate, while the corresponding modes encode spatial or spectral amplitudes. For audio and vibration signals, DMD is typically applied to short-time spectra so that each narrow frequency neighborhood yields a small set of decaying exponentials. This makes DMD attractive for plate responses, where each resonant mode manifests locally as a damped sinusoid in time at a nearby frequency. Related physics-aware learning approaches have recently pursued differentiable modal simulation and inverse rendering [9, 10], while large-scale audio transformers such as AST [11] and PANNs [12] pretrained on AudioSet [13] provide strong global time-frequency representations for impulse responses.

In our Task B study, we combine these two lines of work in a dual-stream grid framework. The global stream uses a pretrained AST backbone with a log-energy descriptor to summarize the full impulse response. The spectral stream operates on a 10 000-bin log-frequency grid aligned with the challenge evaluation band and encodes four local channels: log magnitude, normalized DMD decay, normalized DMD gain, and a DMD validity mask. On the DMD side, we developed a band-wise pipeline tailored to densely modal plate data: the unnormalized displacement response is resampled to 48 kHz, analyzed with a Kaiser-windowed STFT, and processed with rank-3 DMD in local spectral neighborhoods; the resulting decay and amplitude cues are mapped to the same log grid used for ground-truth rasterization and z-score normalized jointly with the training targets. This front-end is precomputed once per dataset split and fed to a one-dimensional convolutional encoder rather than used as a standalone estimator. The fused per-bin representation drives three convolutional heads for mode activity, normalized decay, and normalized gain; at inference, active bins are selected by thresholding the activity head and each active bin exports one mode triplet. Transformer-style fusion of global embeddings with local spectral grids follows the broader trend of combining pretrained audio backbones with structured time-frequency features [14, 15].

2. PROPOSED ALGORITHM

2.1. Problem Formulation and Evaluation Band

For each plate, let the ground-truth modal set contain M modes with parameters $(f_{0,m}, \sigma_m, b_m)_{m=1}^{M}$. The model must output an estimated set of $\tilde{M}$ modes with parameters $(\hat{f}_{0,i}, \hat{\sigma}_i, \hat{b}_i)_{i=1}^{\tilde{M}}$. Following the challenge definition, analysis is restricted to a user-specified band $[f_{\min}, f_{\max}]$. In our experiments we use

$$f_{\min} = 20\,\text{Hz}, \qquad f_{\max} = 10\,000\,\text{Hz}. \tag{1}$$

Modes outside this band are ignored during training rasterization and evaluation. The official ranking metric combines clipped per-mode relative errors with a mode-count term:

$$\begin{aligned} \text{RE}_0 &= \frac{\text{RE}_\Omega + \text{RE}_\sigma + \text{RE}_b}{3}, \\ \text{RE} &= \text{RE}_0 + \lambda \min\left(1, \frac{|\Delta M|}{M}\right), \end{aligned} \tag{2}$$

where $\Delta M = |M - \tilde{M}|$ and $\lambda = 1$ by default. Actual and estimated modes are matched by Hungarian assignment on log-frequency distance.

2.2. Preprocessing

Each sample is loaded from the official `.npz` file. Band-wise DMD features are computed from the unnormalized displacement response `ir`. For the AST branch and log-magnitude grid, the `.npz` response is scaled by the provided `normalization_factor` before feature extraction. Two complementary input representations are then constructed.

The global AST branch converts the waveform to a log-mel filterbank with target length 256 frames and applies an amplitude gain of 10^8 to preserve displacement-scale information. A scalar log-energy feature is appended to the AST class token embedding. The spectral grid branch computes the log magnitude of the real FFT and resamples it onto log-spaced grid centers

$$c_k = \mathcal{G}(f_{\min}, f_{\max}, K), \qquad k = 1, \ldots, K, \tag{3}$$

where $\mathcal{G}(\cdot)$ denotes a logarithmic spacing function that generates K geometrically distributed points between the start and end frequencies, and $K = 10\,000$ is the total number of frequency bins.

Ground-truth modes from `random_IR_modes_*.csv` are mapped to the nearest log bin. If multiple modes fall into the same bin, the mode with the largest absolute gain is kept. This yields three grid targets for each sample: an activity mask $a_k \in \{0, 1\}$, a raw decay σ_k, and a raw gain b_k. Before z-score normalization, the decay is transformed with the natural logarithm and the gain magnitude with the base-10 logarithm, denoted as $y_k = \ln(\sigma_k)$ and $z_k = \log_{10}|b_k|$, respectively. The normalized targets are then computed as:

$$\sigma_k^{\text{norm}} = \frac{y_k - \mu_y}{s_y}, \qquad b_k^{\text{norm}} = \frac{z_k - \mu_z}{s_z}, \tag{4}$$

where μ_y, μ_z denote the training-set means, and s_y, s_z denote the sample standard deviations of the corresponding log-transformed domains.

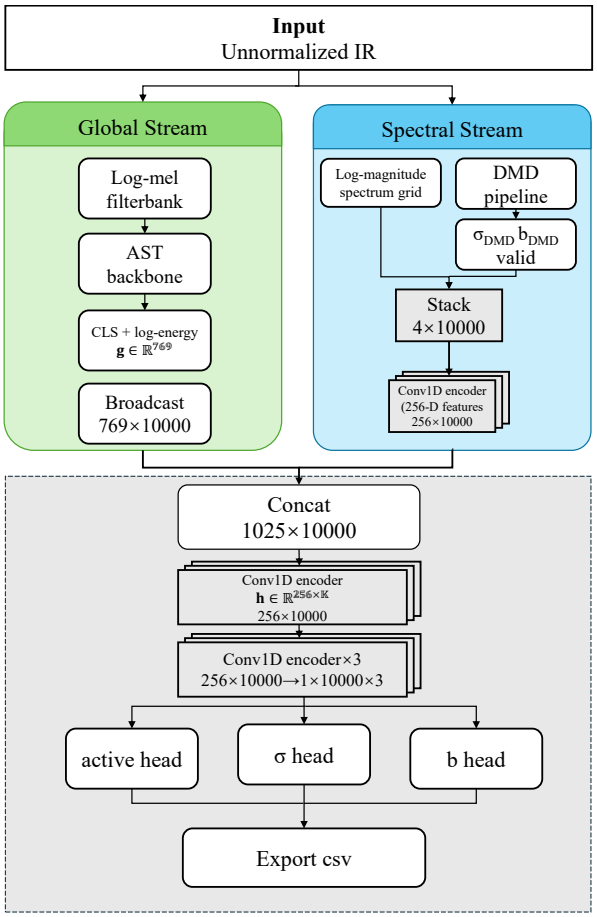


Figure 1: Overall architecture of the proposed dual-stream AST grid framework for Task B.

2.3. Dual-Stream AST Grid Network

The complete architecture is shown in Fig. 1. The proposed network contains two streams fused at each of the 10 000 frequency bins, as shown in Table 1. The global stream uses an Audio Spectrogram Transformer (AST) backbone [11] initialized from AudioSet pretraining [13]. The local stream applies a three-layer Conv1D encoder to the four-channel log-magnitude and DMD grid. The encoder output is upsampled to length K and concatenated with the global embedding broadcast across bins. Two convolution layers with ReLU and dropout produce a hidden map $\mathbf{h} \in \mathbb{R}^{256 \times K}$, followed by three 1×1 convolutional heads for activity, σ, and gain.

During inference, the raw output logit l_k of the activity head is converted to a continuous presence probability $p_k = \text{sigmoid}(l_k)$. Active bins are then identified by applying a decision threshold $\tau = 0.3$:

$$\hat{a}_k = \mathbb{I}[p_k \geq \tau], \qquad \hat{f}_{0,k} = c_k, \tag{5}$$

where $\mathbb{I}[\cdot]$ denotes the indicator function. For each detected active bin ($\hat{a}_k = 1$), the corresponding physical decay and gain parameters are recovered by inverting the z-score and logarithmic transformations formulated in Eq. (4).

Table 1: Main components of the proposed Task B system.

Component	Symbol / Output	Description
Global AST stream	$\mathbf{g} \in \mathbb{R}^{768+1}$	Pretrained AST CLS embedding plus log-energy feature from the impulse response
Spectral grid stream	$\mathbf{S} \in \mathbb{R}^{4\times K}$	Log-magnitude spectrum and DMD channels resampled to 10 000 bins
Per-bin fusion	$\mathbf{h}_k$	Concatenation of local spectral features and broadcast global embedding
Active head	a_k	Binary mode presence logit at each grid bin
Regression heads	$\sigma_k^{\mathrm{norm}}, b_k^{\mathrm{norm}}$	Normalized decay and gain predictions at each bin

2.4. DMD Feature Extraction

To inject physics-inspired local dynamics into the spectral encoder, we extract band-wise Dynamic Mode Decomposition (DMD) features following a pipeline adapted from the challenge reference script [5, 7]. The unnormalized displacement response is first resampled to $48\,\mathrm{kHz}$ and transformed using a Short-Time Fourier Transform (STFT). Specifically, we apply a Kaiser window ($\beta = 1.9\pi$) with a frame size of 2048 samples and a hop size of 512 samples (an overlap factor of 4).

To capture localized modal behavior, the resulting complex spectrogram is analyzed across sliding frequency neighborhoods. For each target frequency bin, we extract a local sub-matrix spanning ± 15 bins (a total bandwidth of 31 frequency bins). To avoid boundary artifacts, the first and last three temporal frames of this sub-matrix are discarded. Exact DMD is then performed on this cropped local neighborhood: we compute the Singular Value Decomposition (SVD) of the snapshot matrices and strictly truncate the subspace rank to $R = 3$.

The discrete-time eigenvalues λ and corresponding complex amplitudes b obtained from the reduced DMD operator are subsequently converted to continuous-time physical parameters. The decay constant is computed as $\sigma = -\mathrm{Re}(\ln(\lambda)/\Delta t)$, where Δt is the time step between frames, and the gain magnitude is taken as $|b|$. Finally, for each valid band, the extracted decay and gain parameters are mapped to the nearest center frequency on the 10 000-bin log grid. These parameters are log-transformed and z-score normalized using the identical training-set statistics formulated in Eq. (4). The resulting DMD channels, along with a binary validity mask, are concatenated with the log-magnitude spectrum to form the four-channel input for the convolutional encoder. All DMD tensors are precomputed per dataset split and stored on disk.

2.5. Loss Function

To cope with the extreme class imbalance (active bins typically constitute less than 10% of all bins) and to reduce sensitivity to outlier predictions, we replace the standard binary cross-entropy with Focal Loss for activity detection, and substitute the mean squared error with Huber Loss for regression. Additionally, we introduce a frequency-dependent weighting that emphasizes lower frequencies, which are perceptually and physically more important.

Let a_k be the ground-truth activity label (0 or 1) and $\hat{p}_k = \mathrm{sigmoid}(l_k)$ the predicted probability. The Focal Loss for activity detection is defined as:

$$\begin{aligned}\mathcal{L}_{\mathrm{focal}} = -\frac{1}{K}\sum_{k=1}^{K}\Big[&\alpha(1-\hat{p}_k)^{\gamma}\log(\hat{p}_k)a_k \\ &+ (1-\alpha)\hat{p}_k^{\gamma}\log(1-\hat{p}_k)(1-a_k)\Big],\end{aligned} \tag{6}$$

where we set $\gamma = 2$ and $\alpha = 0.35$ (equivalent to a positive weight of 0.35 in the original BCE formulation).

For the regression branches, we compute the Huber loss on the normalized targets σ_k^{norm} and b_k^{norm} only at bins where $a_k = 1$:

$$\mathcal{L}_{\mathrm{huber}}(x,y) = \begin{cases} \frac{1}{2}(x-y)^2, & |x-y| \le \delta, \\ \delta(|x-y| - \frac{1}{2}\delta), & \text{otherwise}, \end{cases} \tag{7}$$

with $\delta = 1.0$. The total regression loss is:

$$\begin{aligned}\mathcal{L}_{\mathrm{reg}} = w_\sigma \frac{1}{N_{\mathrm{act}}}\sum_{k:a_k=1}\mathcal{L}_{\mathrm{huber}}(\hat{\sigma}_k^{\mathrm{norm}}, \sigma_k^{\mathrm{norm}}) \\ + w_b \frac{1}{N_{\mathrm{act}}}\sum_{k:a_k=1}\mathcal{L}_{\mathrm{huber}}(\hat{b}_k^{\mathrm{norm}}, b_k^{\mathrm{norm}}),\end{aligned} \tag{8}$$

where $N_{\mathrm{act}} = \sum_k a_k$, and we set $w_\sigma = w_b = 3.0$ to increase the importance of accurate decay and gain estimation.

Finally, we apply a frequency-dependent weight v_k to the regression loss to encourage the model to focus on lower frequencies:

$$v_k = \exp\left(-\frac{c_k}{500}\right) \Big/ \left(\frac{1}{K}\sum_{j=1}^{K}\exp\left(-\frac{c_j}{500}\right)\right), \tag{9}$$

where c_k is the center frequency of bin k in Hz. The weighted regression loss becomes:

$$\begin{aligned}\mathcal{L}_{\mathrm{reg}}^{\mathrm{w}} = w_\sigma \frac{1}{N_{\mathrm{act}}}\sum_{k:a_k=1} v_k\mathcal{L}_{\mathrm{huber}}(\hat{\sigma}_k^{\mathrm{norm}}, \sigma_k^{\mathrm{norm}}) \\ + w_b \frac{1}{N_{\mathrm{act}}}\sum_{k:a_k=1} v_k\mathcal{L}_{\mathrm{huber}}(\hat{b}_k^{\mathrm{norm}}, b_k^{\mathrm{norm}}).\end{aligned} \tag{10}$$

The overall training objective is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{focal}} + \mathcal{L}_{\mathrm{reg}}^{\mathrm{w}}. \tag{11}$$

3. EXPERIMENTAL SETUP

We train on 10 000 synthetic one-second displacement impulse responses and evaluate on a held-out validation set of 1000 responses generated by the official data-generation pipeline. The

proposed system uses a 10 000-bin log grid over the analysis band in Eq. (1) and precomputed DMD caches for the training and validation splits. The AST backbone is initialized from AudioSet pretraining; it is then fine-tuned jointly with the grid heads using AdamW [16] with weight decay 10^{-4}, backbone learning rate 5×10^{-6}, head learning rate 10^{-4}, and a 3-epoch warmup followed by cosine decay. Training runs for 100 epochs with batch size 4; the checkpoint with lowest validation loss (epoch **92** in our run) is used for inference with activity threshold $\tau = 0.411$ (optimized on the validation set). The regression weights are set to $w_\sigma = w_b = 3.0$, and the Focal Loss parameters are $\gamma = 2$, $\alpha = 0.35$. Frequency weighting is enabled. Data loading uses 2 worker processes.

Experiments were conducted on a Linux server with two Intel Xeon Platinum 8180 processors (28 physical cores per socket, 56 threads per socket, 2.5 GHz base frequency, up to 3.8 GHz turbo) and 1 TB of system memory. The server is equipped with eight NVIDIA GeForce RTX 3090 GPUs (24 GiB GDDR6X each, driver version 560.35). Model training and neural-network inference each occupied a single RTX 3090. Official Task B metric evaluation over the validation set was executed in parallel on the CPU using eight worker processes. Implementations use Python 3 and PyTorch [17].

Inference exports one CSV per plate with columns `f0_ident`, `sigma_ident`, and `gain_ident`. Evaluation uses the official Hungarian-matching implementation with the same frequency band as training.

4. CONCLUSIONS

We presented a dual-stream AST grid framework for Task B modal parameter estimation over 10 000 log-spaced frequency bins. By fusing a pretrained AST embedding with a DMD-augmented log-magnitude spectral encoder, the method addresses densely modal plate responses within a 20 Hz to 10 000 Hz analysis band. All modal parameters are predicted by the three convolutional heads at each active grid bin.

5. REFERENCES

[1] K. Arcas and A. Chaigne, "On the quality of plate reverberation," *Applied Acoustics*, vol. 71, no. 2, pp. 147–156, 2010.

[2] N. H. Fletcher and T. D. Rossing, *The Physics of Musical Instruments*, 2nd ed. New York, NY, USA: Springer, 1998.

[3] J. O. Smith, "Music applications of digital waveguides," CCRMA, Department of Music, Stanford University, Tech. Rep. STAN-M-39, 1987.

[4] S. Bilbao, "Sound synthesis for nonlinear plates," in *Proceedings of the 8th International Digital Audio Effects Conference*, 2005, pp. 243–248.

[5] P. J. Schmid, "Dynamic mode decomposition of numerical and experimental data," *Journal of Fluid Mechanics*, vol. 656, pp. 5–28, 2010.

[6] C. W. Rowley, I. Mezić, S. Shams, S. Back, T. Colonius *et al.*, "Spectral analysis of nonlinear flows," *Journal of Fluid Mechanics*, vol. 641, pp. 115–127, 2009.

[7] J. H. Tu, C. W. Rowley, D. M. Luchtenburg, S. L. Brunton, and J. N. Kutz, "On dynamic mode decomposition: Theory and applications," *Journal of Computational Dynamics*, vol. 1, no. 2, pp. 391–421, 2014.

[8] J. Kutz, S. Brunton, B. Brunton, and J. Proctor, *Dynamic Mode Decomposition: Data-Driven Modeling of Complex Systems*, ser. Other Titles in Applied Mathematics. Society for Industrial and Applied Mathematics, 2016.

[9] R. Diaz and M. Sandler, "Fast differentiable modal simulation of non-linear strings, membranes, and plates," in *Proc. Int. Conf. Digital Audio Effects (DAFx25)*, 2025.

[10] X. Jin, C. Xu, R. Gao, J. Wu, G. Wang, and S. Li, "DiffSound: Differentiable modal sound rendering and inverse rendering for diverse inference tasks," in *Proc. ACM SIGGRAPH Conf. Papers*, 2024.

[11] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio spectrogram transformer," in *Proc. Interspeech*, 2021, pp. 571–575.

[12] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, "PANNs: Large-scale pretrained audio neural networks for audio pattern recognition," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 2880–2894, 2020.

[13] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "AudioSet: An ontology and human-labeled dataset for audio events," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2017, pp. 776–780.

[14] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[15] G. Elbanna, A. Biryukov, N. Scheidwasser-Clow, L. Orlandic, P. Mainar, M. Kegler, P. Beckmann, and M. Cernak, "Hybrid handcrafted and learnable audio representation for analysis of speech under cognitive and physical load," in *Proc. Interspeech*, 2022.

[16] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *CoRR*, vol. abs/1412.6980, 2014.

[17] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.